

EXAM 7 – SPRING 2015

15. (2.25 points)

Given the following forecasted information for a company (\$000):

	<u>2016</u>	<u>2017</u>	<u>2018</u>
Beginning GAAP Equity	100,000	105,000	107,000
Net Income	10,000	12,000	13,000
Minimum Capital Requirement	105,000	107,000	110,000

- Investors demand 8% return on equity.

a. (1.25 points)

Assuming 3% growth in abnormal earnings in perpetuity, calculate the company value as of January 1, 2016.

b. (0.5 point)

Assuming abnormal earnings grow 3% for 3 years beyond the forecast horizon, and then decrease to \$0 for 2022 and all subsequent years, calculate the company value as of January 1, 2016.

c. (0.5 point)

Discuss which of the two scenarios presented above is more realistic.

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EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 15	
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE: B2: Value the equity of a P&C insurer based on its expected future dividends, its free cash flow to equity, or its expected abnormal earnings.
SAMPLE ANSWERS	
Part a: 1.25 points	
<i>Sample Answer 1</i> $K = .08$ ROE = $\frac{2016}{10/100} = .1$ $\frac{2017}{12/105} = .114$ $\frac{2018}{13/107} = .121$ $V = 100,000 + (.1 - .08)(100,000)/1.08 + (.114 - .08)(105,000)/1.08^2 + (.121 - .08)(107,000)/1.08^3 + [(.121 - .08)(107,000)(1.03)/(.08 - .03)]/1.08^3$ $= 100,000 + 8,395 + 71,740$ $= 180,135$ <i>Sample Answer 2</i> $AE_{2016} = 10,000 - 100,000 * 0.08 = 2,000$ $AE_{2017} = 12,000 - 107,000 * 0.08 = 3,600$ $AE_{2018} = 13,000 - 105,000 * 0.08 = 4,440$ $V_0 = BV_0 + \sum AE/(1+k)^i + \text{Terminal Value}$ $= 100,000 + 2,000/1.08 + 3,600/1.08^2 + 4,440/1.08^3 + [4,440*1.03/(0.08-0.03)]/1.08^3$ $= 181,070$	
Part b: 0.5 point	
<i>Sample Answer 1</i> $V_0 = 100,000 + (.1 - .08)(100,000)/1.08 + (.114 - .08)(105,000)/1.08^2 + (.121 - .08)(107,000)/1.08^3 + (.121 - .08)(107,000)(1.03)/1.08^4 + (.121 - .08)(107,000)(1.03)^2/1.08^5 + (.121 - .08)(107,000)(1.03)^3/1.08^6$ $= 100,000 + 8,395 + 3,321 + 3,168 + 3,021$ $= 117,905$ <i>Sample Answer 2</i> $V_0 = 100,000 + 2,000/1.08 + 3,600/1.08^2 + 4,440/1.08^3 + 4,440*1.03/1.08^4 + 4,440*1.03^2/1.08^5 + 4,440*1.03^3/1.08^6$ $= 118,088$	
Part c: 0.5 point	
<i>Sample Answer 1</i> • Scenario in part (b) is more realistic • Abnormal earnings cannot be expected to continue in perpetuity <i>Sample Answer 2</i> (b) is more realistic since maintaining abnormal earnings in perpetuity is not realistic in practice	

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EXAMINERS' REPORT
Overall, candidates performed rather well on this question. Candidates were expected to know how to calculate abnormal earnings and apply them to the abnormal earnings method in the estimation of company value. Candidates lost credit if they based the required shareholder return on the minimum capital requirement rather than the beginning GAAP equity. In addition, many candidates assumed that the abnormal earnings tended towards zero on a linear basis rather than a one-time decrease to \$0. Many candidates also lost points as they failed to include the initial book value in the final calculation.
Part a
Candidates were expected to apply the abnormal earnings method to provide an estimate of company value. In order to receive full credit, candidates were required to show the appropriate calculation of abnormal earnings discounted at a rate of 8%. In addition, candidates were required to calculate the appropriate discounted terminal value since the abnormal earnings were projected to go into perpetuity. Many candidates performed very well. A common mistake was estimating the abnormal earnings based on the minimal capital requirement rather than beginning GAAP equity. Many candidates also failed to add the original book value to the value contemplated by the abnormal earnings.
Part b
Part b is an extension of part a, and again, many candidates performed well. In order to receive full credit, candidates were expected to adjust the 2018 abnormal earnings by 3% per annum and discount it using an 8% discount rate. Candidates who did not receive full credit typically assumed that the abnormal earnings would gradually trend to zero linearly.
Part c
Candidates performed very well on this question part. Candidates were expected to understand that it is not realistic to assume that abnormal earning to continue into perpetuity.